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NON-TRANSITORY COMPUTER READABLE STORAGE MEDIUM AND INFORMATION PROCESSING APPARATUS

Abstract

A non-transitory computer readable storage medium stores a program set including a resident first program and a non-resident second program. The second program, when executed by a processor, causes an information processing apparatus to perform second operations including: transmitting, to the image processing device, an execution instruction for causing a job related to image processing to be executed; and displaying a setting screen configured to receive a setting related to the first program. The first program, when executed by the processor, causes the information processing apparatus to perform first operations including: communicating, in accordance with the setting received on the setting screen displayed in the second operations, with the image processing device; acquiring information related to the image processing device through a communication with the image processing device; and displaying notification information based on the acquired information related to the image processing device.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority from Japanese Patent Application No. 2024-034175 filed on Mar. 6, 2024. The entire content of the priority application is incorporated herein by reference.

BACKGROUND ART

[0002] The technical field disclosed in the present specification relates to a non-transitory computer readable storage medium storing a program set, and an information processing apparatus, and particularly, the program set includes a resident program configured to acquire information related to an image processing device.

[0003] As an example of a resident program that functions as a resident process, which is resident after an information processing apparatus is activated and continues processing, there is a program for communicating with an image processing device to be monitored. For example, a status monitor caused to function by a resident program periodically communicates with a printer, acquires a status of the printer, and generates a notification of information indicating the status as necessary.

DESCRIPTION

[0004] The resident program is resident after the information processing apparatus is activated, and the resident program continues to consume resources such as a memory. Therefore, there is room for improvement in regard to resource consumption of the resident program.

[0005] In an aspect of the present disclosure, there is provided a non-transitory computer readable storage medium storing a program set, the program set including a first program and a second program, the first program being a resident program, the second program being a non-resident program, wherein the second program, when executed by a processor of an information processing apparatus, causes the information processing apparatus to perform second operations, the second operations including: transmitting, to an image processing device via a communication device of the information processing apparatus, an execution instruction for causing a job related to image processing to be executed; and displaying a setting screen configured to receive a setting related to the first program, and wherein the first program, when executed by the processor, causes the information processing apparatus to perform first operations, the first operations including: communicating, in accordance with the setting received on the setting screen displayed in the second operations, with the image processing device via the communication device; acquiring information related to the image processing device through a communication with the image processing device; and displaying notification information based on the information related to the image processing device acquired in the acquiring the information related to the image processing device.

[0006] According to the technique disclosed in the present specification, separate from the resident first program for acquiring information related to the image processing device, the non-resident second program for transmitting an image processing execution instruction is provided to include the setting screen for the first program. Consequently, an amount of resources consumed by the first program can be reduced, whereby resources for displaying the setting screen for making a setting related to the first program can be efficiently used as compared with a case in which the setting screen is included in the first program.

[0007] An information processing apparatus configured to achieve the operations by the above device is also novel and useful.

Description

[0008] According to the technique disclosed in the present specification, it is possible to achieve a technique of efficiently using resources of an information processing apparatus when a resident program is used.
[0009] FIG. **1** is a schematic diagram of an information processing system according to a first embodiment.
[0010] FIG. **2** is a diagram illustrating a display of an operation display unit in a state in which a first status information screen is displayed as a pop-up.
[0011] FIG. **3**A is a diagram illustrating a second status information screen displayed as a pop-up on the display of the operation display unit; FIG. **3**B is a diagram illustrating a third status information screen displayed as a pop-up on the display of the operation display unit; and FIG. **3**C is a diagram illustrating an offering information screen displayed as a pop-up on the display of the operation display unit.
[0012] FIG. **4** is a sequence diagram illustrating an example of a procedure for display processing of status information, offering information, and a notification list, implemented by a device monitoring program of an information processing apparatus, including operations of a device and a server.
[0013] FIG. **5** is an operation diagram illustrating the display of the operation display unit in the state in which the first status information screen is displayed as a pop-up.
[0014] FIG. **6** is a diagram illustrating an example of a notification list screen.
[0015] FIG. **7** is a sequence diagram illustrating an example of a procedure for various setting processing implemented by the device monitoring program and a device control program of the information processing apparatus, including operations of the device.
[0016] FIG. **8** is a diagram illustrating an example of the display in a state in which a box and an activation box are displayed as pop-ups.
[0017] FIG. **9** is a diagram illustrating an example of the display in a state in which a second home screen displayed by the device control program is displayed.
[0018] FIG. **10** is a diagram illustrating an example of the display in a state in which a setting screen is displayed.
[0019] FIG. **11** is a diagram illustrating an example of the display in a state in which a device list screen is displayed.
[0020] FIG. **12** is a diagram illustrating an example of the display in a state in which a device addition screen is displayed.
First Embodiment
[0021] Hereinafter, a specific embodiment of an information processing system according to the present embodiment will be described in detail with reference to the accompanying drawings. In the present embodiment, the present disclosure is applied to the information processing system including a printer and an information processing apparatus that can communicate with each other.
[0022] That is, as illustrated in FIG. **1**, an information processing system **100** according to the present embodiment includes four devices **1**A, **1**B, **1**C, and **1**D and an information processing apparatus **200**, which are communicatively connected to each other. In the following description, when there is no need to distinguish between the devices **1**A, **1**B, **1**C, and **1**D, the subscripts for distinguishing A to D will be omitted and the devices will be described as a device **1**. The device is an example of an image processing device.
[0023] The device **1** includes a printer **1**A capable of printing on a medium to be printed, and an image reading device **1**B capable of reading an image recorded on a medium to be read. In the present embodiment, the printer **1**A is exemplified as an inkjet printer. As a modification, the printer **1**A may be a laser printer, a label printer, a copier, or a multi function device. The

information processing apparatus **200** generates and edits image data to be printed by the printer **1**A, transmits a print execution instruction and the image data to the printer **1**A, and receives status information from the printer **1**A.

[0024] The information processing apparatus **200** transmits a reading execution instruction to the image reading device **1**B, receives read data from the image reading device **1**B, and receives status information from the image reading device **1**B. The image reading device **1**B may be a copier, a multi function device, or the like.

[0025] The information processing apparatus **200** is, for example, a smartphone, a personal computer, or a tablet computer. The number of information processing apparatus **200** constituting the information processing system **100** is not limited to one. A plurality of information processing apparatus may be provided.

[0026] As illustrated in FIG. **1**, in the present embodiment, the printer **1**A includes a controller **11**, a print unit **12**, an operation panel **13**, a network interface **14**, and a storage unit **15**. The controller **11** includes a CPU and memories such as a flash ROM and a RAM, and controls each component of the printer **1**A.

[0027] The controller **11** is a general term for hardware used to control the printer **1**A, such as the CPU. Specifically, the controller **11** may also include an application specific integrated circuit (ASIC) and the like.

[0028] The print unit **12** is configured to print an image on a medium to be printed based on the image data transmitted from the information processing apparatus **200**. In the present embodiment, an inkjet printing method is adopted as an image forming method of the print unit **12**. The print unit **12** includes a print head that prints an image on a medium. Further, the print unit **12** may also be referred to as a print engine. The image forming method of the print unit **12** may be a thermal printing method or an electrophotographic method.

[0029] The operation panel **13** is, for example, a touch panel, receives an input from a user, and displays information. The operation panel **13** may include various display lamps, buttons, and the like.

[0030] The network interface **14** is hardware for communicating with the information processing apparatus **200**.

[0031] The storage unit **15** of the printer **1**A stores a program for an embedded web server (hereinafter, referred to as "EWS" (abbreviation of embedded web server)) **16**. Further, the storage unit **15** has a data storage area **17**, and the data storage area **17** stores, for example, capability information on the printer **1**A, connection information for connecting to the printer **1**A, status information on the printer **1**A, various setting information set in the printer **1**A, consumable remaining amount information on the printer **1**A, printed sheet number information on the printer **1**A, and consumable support-related information. For example, the information processing apparatus **200** can access various information stored in the data storage area **17** via the EWS **16**. The printer **1**A manages a print progress status for each transmission source of a job. In the present embodiment, the consumable is an ink cartridge that contains ink consumed in print processing of a printer, and the remaining amount information is information related to a remaining amount of ink contained in the ink cartridge.

[0032] The devices **1**B to **1**D also include an EWS similar to the EWS **16** of the printer **1**A, and a storage unit (both not illustrated).

[0033] In the present embodiment, as illustrated in FIG. **1**, the information processing apparatus **200** includes a desktop personal computer (PC) **2** and an input and output device **25**. The desktop personal computer (PC) **2** includes a CPU **21**, a ROM **22**, a RAM **23**, a non-volatile memory **24**, an interface **26**, and a network interface **27**. The ROM **22** stores various programs and the like for the information processing apparatus **200**. The RAM **23** is used as a work area when various processing is executed, or as a storage area for temporarily storing data. The non-volatile memory **24** is, for example, a hard disk drive (HDD), a solid state drive (SSD), a flash memory, or the like,

and stores various programs and data.
[0034] The CPU **21** executes various processing according to programs loaded from the ROM **22** to the non-volatile memory **24**.
[0035] The input and output device **25** is connected to the interface **26**, and includes an input device and an output device. The input device includes a keyboard **40**, a mouse **41**, and the like. The output device includes a display **42**. The display **42** is, for example, a liquid crystal display or an organic EL display. The input and output device **25** may be a touch panel that can be operated by touch. The information processing apparatus **200** may be a notebook PC or a tablet.
[0036] The network interface **27** is hardware for communicating with each device **1**. The network interface **27** is an example of a communication unit.
[0037] In the information processing system **100** according to the present embodiment, the information processing apparatus **200** and each device **1** perform wireless communication conforming to a Wi-Fi (registered trademark) standard via a common access point **3**. That is, the network interfaces **14** and **27** are both wireless interfaces that enable wireless communication based on the Wi-Fi (registered trademark) standard. The network interfaces **14** and **27** may be wired interfaces such as wired local area network (LAN) interfaces.
[0038] The communication mode according to the present embodiment is an example, and is not limited to the above mode. For example, the information processing apparatus **200** and each device **1** may communicate with each other by wireless communication through a direct connection without going through the access point **3**, or by wired communication using an interface such as USB.
[0039] The access point **3** is connected to a server **5** provided by a manufacturer, a sales company, or the like of the device **1** via an Internet **4**. Therefore, the information processing apparatus **200** can download a manual or the like of the device **1** from the server **5** via the Internet **4**. In addition to the manual, offering information such as service information and bargain information related to a consumable of the device **1** can be downloaded from the server **5**.
[0040] Examples of services related to the consumable include an automatic consumable delivery service and a subscription service. The automatic consumable delivery service is a service that automatically delivers a consumable to a user of the printer based on consumable remaining amount information. The subscription service is a service that charges a fixed amount for up to a predetermined number of printed sheets of the printer, and automatically delivers the consumable to the user of the printer based on the consumable remaining amount information.
[0041] As illustrated in FIG. **1**, the non-volatile memory **24** of the information processing apparatus **200** stores an operating system (OS) **31**, an application program (app) **32**, a device monitoring program **33**, and a device control program **34**. Further, the non-volatile memory **24** includes a data storage unit **35**.
[0042] The OS **31** is, for example, Windows (registered trademark), macOS (registered trademark), Linux (registered trademark), iOS (registered trademark), or Android (registered trademark). In the present embodiment, the OS **31** is Windows.
[0043] The app **32** is, for example, an image editing program having a function of receiving a user instruction via the interface **26**, a function of displaying an image on the display **42**, and a function of editing and saving the image. The app **32** may be a document editing program, a spreadsheet program, or the like.
[0044] The device monitoring program **33** is a resident program, and is a program for implementing a function of periodically communicating with each device **1** via the network interface **27** and acquiring various status information and the like from each device **1**. Further, the device monitoring program **33** is a program for implementing a function of periodically communicating with the server **5** via the network interface **27** and acquiring various information from the server **5**.
[0045] The resident program is, for example, a program that is automatically activated when the

OS is activated and performs predetermined processing in a background during activation of the OS. The device monitoring program **33** is an example of a first program.
[0046] Specifically, the device monitoring program **33** acquires, from the printer **1**A via the network interface **27**, the capability information on the printer **1**A, the connection information for connecting to the printer **1**A, the status information on the printer **1**A, the various setting information set in the printer **1**A, the consumable remaining amount information on the printer **1**A, the printed sheet number information on the printer **1**A, the consumable support-related information, information on the print progress status, and the like, and stores the acquired information in the data storage unit **35**.
[0047] Further, the device monitoring program **33** acquires, from the image reading device **1**B via the network interface **27**, capability information on the image reading device **1**B, connection information for connecting to the image reading device **1**B, status information on the image reading device **1**B, various setting information set in the image reading device **1**B, information related to a read progress status, and the like, and stores the acquired information in the data storage unit **35**.
[0048] Further, the device monitoring program **33** acquires, via the network interface **27**, the offering information related to the device **1** provided by the manufacturer or the sales company of the device **1**, and stores the acquired offering information in the data storage unit **35**. The offering information includes consumable-related service information and bargain information, maintenance information, notification of a campaign, a latest model, and introductions of convenient functions. Further, the offering information may include link information to the server **5**. The offering information is an example of service-related information.
[0049] Then, the device monitoring program **33** can read the status information, the offering information, and the like of the device **1** stored in the data storage unit **35**, and display the read information as a pop-up on the display **42**.
[0050] The device control program **34** has a function of causing the printer **1**A, which is set in a model name display field **73** described later, to print an image edited or stored by the app **32**. That is, the device control program **34** receives a selection of an image to be printed, and transmits print data based on the selected image to the printer **1**A, thereby causing the printer **1**A to perform printing based on the print data. This function is an example of transmission processing. The status information, the consumable remaining amount information, and the like on the selected printer **1**A can be read from the data storage unit **35** and displayed on the display **42**. The device control program **34** is, for example, a non-resident program that is activated when activated and operated by the user. The device control program **34** is an example of a second program.
[0051] The device control program **34** instructs the selected image reading device **1**B to read, receives image information read by the selected image reading device **1**B, and stores the image information in the data storage unit **35**. This function is an example of transmission processing. Further, information related to a read progress status of the selected image reading device **1**B can be read from the data storage unit **35** and then displayed on the display **42**.
[0052] The data storage unit **35** is provided with a device table **35**A. The device table **35**A stores model names, model numbers, and the like of all the devices **1** to be monitored by the device monitoring program **33**.
[0053] FIG. **2** is a diagram illustrating an example of a display state of the display **42** in a state in which a first status information screen **50** by the device monitoring program **33** is displayed as a pop-up. That is, on a first home screen **42**A of the information processing apparatus **200**, an icon **43** for the user to activate the device control program **34** is displayed on a left side of the display **42**, and a taskbar **44** is displayed on a lower side of the display **42**. A box display button **45** is displayed on a right side of the taskbar **44**. The icon **43** for the device control program **34** is not required to be displayed on the first home screen **42**A, and the device control program **34** may be activated by an operation on a menu screen of the information processing apparatus **200** (not illustrated).
[0054] The first home screen **42**A is displayed on the display **42** by the OS, and on the first home

screen **42**A, the icon **43**, the taskbar **44**, the box display button **45**, and the like are displayed by the OS.

[0055] On the first home screen **42**A displayed on the display **42**, a first status information screen **50** related to a consumable, for example, an ink remaining amount, is displayed as a pop-up. The first status information screen **50** includes a first character string display field **50**A in which a character string such as “replace ink soon” is displayed as status information, a second character string display field **50**B in which a character string such as “access website containing support information” is displayed and which is linked to an access destination of the server **5**, a model name display field **50**C in which a model name of the device **1** is displayed, an ink remaining amount display field **50**D in which an ink remaining amount is displayed, a sheet remaining amount display field **50**E in which a sheet remaining amount is displayed, an offering information display field **50**F in which offering information or the like related to the status information displayed in the first character string display field **50**A is displayed, a notification list button display field **50**G in which a notification list button for displaying a notification list screen is displayed, and the like.

[0056] The model name display field **50**C displays a model name and an IP address of a corresponding device. Instead of the IP address, other identification information that can uniquely identify the device may be used. As other identification information, for example, a MAC address, a serial number, or the like can be used.

[0057] Further, on the first home screen **42**A, a second status information screen **51** not related to a consumable, for example, ink, as illustrated in FIG. **3**A is displayed as a pop-up by the device monitoring program **33**. The second status information screen **51** includes a model name display field **51**A in which the model name of the device **1** is displayed, and a second character string display field **51**B in which a character string such as “no sheet” is displayed as status information. When the character string “no sheet” is displayed in the second character string display field **51**B, the information processing apparatus **200** acquires status information indicating that there is no sheet from the printer **1**.

[0058] Further, on the first home screen **42**A, a third status information screen **52** not related to a consumable, for example, ink, as illustrated in FIG. **3**B is displayed as a pop-up by the device monitoring program **33**. The third status information screen **52** includes a model name display field **52**A in which the model name of the device **1** is displayed, and a progress status display field **52**B in which, for example, a scan progress status of the image reading device **1**B is displayed as status information.

[0059] Further, on the first home screen **42**A, an offering information screen **53** that displays offering information or the like as illustrated in FIG. **3**C, is displayed as a pop-up by the device monitoring program **33**. The offering information screen **53** includes an offering information display field **53**A in which offering information is displayed, a notification list button display field **53**B in which a notification list button for displaying a notification list screen is displayed, and the like.

[0060] The information processing apparatus **200** may be configured such that on the first home screen **42**A, the first status information screen **50** that displays the status information related to the consumable, the second status information screen **51** and the third status information screen **52** that display the status information not related to the consumable, and the offering information screen **53** that displays the offering information are displayed as pop-ups not to overlap each other, or these information screens **50** to **52** are overlapped and displayed as pop-ups.

[0061] Further, when the information processing apparatus **200** is configured such that a plurality of screens are overlapped and displayed as pop-ups, the information processing apparatus **200** may be configured such that a screen that displays new information is displayed as a pop-up in a foreground. In this case, since the screen that displays the new information is displayed as a pop-up in the foreground, the user can always keep an eye on a latest status of the device and latest

information on the device, and acquire the status and the information, enabling a quick response.
Display of Status Information, Offering Information, and Notification List
[0062] FIG. **4** is a sequence diagram illustrating an example of a procedure for displaying the status information, the offering information, and the notification list implemented by the device monitoring program **33** of the information processing apparatus **200**, including operations of the device **1** and the server **5**.
[0063] Each processing step in the present embodiment represents processing of the CPU **21** according to an instruction described in a program such as the device monitoring program **33** of the information processing apparatus **200**. The processing by the CPU **21** also includes a hardware control using an API of the OS. In the present specification, a detail description of the OS is omitted, and an operation of each program is described. In addition, in the present specification, for convenience, processing primarily performed by the CPU **21** may be described as being primarily performed by a program or a device.
Display of Status Information and Offering Information
[0064] In FIG. **4**, first, in a case in which an error occurs during operation, for example, in a case in which the device **1** is the printer **1**A, when a consumable such as ink runs out or an error such as a sheet jam occurs during printing, the device **1** transmits status information indicating error content to the information processing apparatus **200** (procedure **11** (hereinafter, referred to as T**11**)). Therefore, the information processing apparatus **200** that receives the status information stores the received status information in the data storage unit **35** (T**12**).
[0065] In the information processing apparatus **200**, a query for status information is executed on each device **1** connected to the information processing apparatus **200**, that is, on each device **1** to be monitored by the device monitoring program **33**, that is, on the device **1** having the model name stored in the device table **35**A. In other words, the information processing apparatus **200**, that is, the device monitoring program **33** queries each device **1** at regular intervals, and a so-called polling is executed (T**13**). A polling interval for each device **1** may be constant, or may be varied based on, for example, a frequency of use of the device **1**. Further, the polling interval may be a polling interval set by the user on a setting screen **80** described later.
[0066] Each device **1** that receives the query from the information processing apparatus **200** transmits the status information to the information processing apparatus **200** (T**14**). Therefore, the device monitoring program **33** that receives the status information stores the received status information in the data storage unit **35** (T**15**). The processing in T**14** is an example of the acquisition processing.
[0067] When the status information is stored in the data storage unit **35** in T**12** or T**15**, the device monitoring program **33** determines whether the stored status information is status information related to a consumable such as ink provided by the manufacturer or the sales company of the device **1**. If the device monitoring program **33** determines that the status information is status information related to the consumable, as illustrated in FIG. **2**, the device monitoring program **33** displays the first status information screen **50** as a pop-up on the first home screen **42**A based on content of the status information (T**16**).
[0068] Further, the device monitoring program **33** searches for offering information stored in the data storage unit **35** based on the content of the status information and the model name of the device **1**, specifically, for example, warning information “replace ink soon” displayed in the first character string display field **50**A and a model name “DCP-XXXDW” displayed in the model name display field **50**C, and extracts the offering information stored in association with the warning information and the model name. Then, the information processing apparatus **200** displays the extracted offering information in the offering information display field **50**F on the first status information screen **50**.
[0069] As described above, in the present embodiment, on the first status information screen **50** based on the status information related to the consumable among the status information transmitted

from all the devices **1** connected to the information processing apparatus **200**, that is, the respective devices **1** to be monitored by the device monitoring program **33**, the offering information related to the warning information and the model name is also displayed in the offering information display field **50**F, and therefore, by viewing the content displayed as a pop-up, the user can recognize that there is a device **1** having a predetermined status, or that there is offering information related to the status, thereby improving user convenience. Further, an improvement in user ability to cope with the status information can also be expected.

[0070] Next, the device monitoring program **33** determines whether the status information stored in the data storage unit **35** is status information related to a consumable such as ink provided by the manufacturer or the sales company of the device **1**. If the device monitoring program **33** determines that the status information is not status information related to the consumable provided by the manufacturer or the sales company of the device **1**, such as running out of print sheet, the device monitoring program **33** displays the second status information screen **51** as illustrated in FIG. **3**A or the third status information screen **52** as illustrated in FIG. **3**B as pop-ups on the first home screen **42**A based on the content of the status information (T**17**). The second status information screen **51** is a screen indicating that there is no sheet in the device, and the third status information screen **52** is a screen indicating that scan processing is being executed in the device.

[0071] When the device monitoring program **33** receives the same status information from the device **1** while displaying a screen based on certain status information from the device **1** as a pop-up, the device monitoring program **33** may omit further displaying a screen based on the received status information as a pop-up. The processing in T**16** and T**17** are examples of the notification processing.

[0072] Next, as illustrated in FIG. **2**, in a state in which the device monitoring program **33** displays the first status information screen **50** as a pop-up, when a list button **50**H portion displayed on a right side of the model name display field **50**C is clicked with the mouse **41** to instruct a display of pull-down (T**18**), the device monitoring program **33** displays a menu **50**J below the model name display field **50**C as illustrated in FIG. **5** (T**19**). The list menu displays model names and IP addresses of all the devices **1** to be monitored by the device monitoring program **33**, that is, model names and IP addresses of the devices stored in the device table **35**A, respectively.

[0073] At this time, the model name display field **50**C and the menu **50**J each display a status mark **50**K corresponding to the model name displayed therein. The status mark **50**K is displayed based on the status information stored in the data storage unit **35**.

[0074] Here, when the user clicks and selects a new model name portion displayed in the menu **50**J with the mouse **41** (T**20**), the device monitoring program **33** displays another model name newly selected by the user in the model name display field **50**C, reads the status information and the offering information corresponding to the model name newly selected by the user from the data storage unit **35**, and displays the status information and the offering information as the first status information screen **50** (T**21**).

[0075] At this time, even if the user selects another model name displayed in the menu **50**J by the device monitoring program **33** in the processing in T**20**, the device selected in the device control program **34** is not changed, and for example, the model name displayed in a model name display field **73** on the second home screen **70** illustrated in FIG. **9** described later is not changed.

[0076] The device monitoring program **33** displays the first status information screen **50** to notify a device to be monitored, and the user can switch the model name display field **50**C to check a status of another device. On the other hand, the device control program **34** selects a device to be controlled, and when the same device is continuously used, there is no need to reselect the device to be controlled. Therefore, if the device controlled by the device control program **34** is switched in link with switching of the device being monitored by the device monitoring program **33**, the user who continues to use the same device needs to restore the device to be controlled. Therefore, it is more convenient for the user when the device **1** set in the model name display field **50**C and the

device **1** set in the model name display field **73** on the second home screen **70** are not linked with each other, as in the present embodiment.
[0077] Further, in the present embodiment, even for another model name displayed in the menu **50**J, by clicking and selecting another model name with the mouse **41**, status information and offering information corresponding to the another model name can be displayed as the first status information screen **50**, and thus latest status information and offering information on the device **1** required by the user can be provided, improving user convenience.
[0078] Further, in the present embodiment, the model name display field **50**C and the menu **50**J each display the status mark **50**K corresponding to the model name displayed therein, which assists the user in selecting another model name displayed in the menu **50**J or the like, thereby improving convenience. Further, the device monitoring program **33** queries the server **5** connected to the information processing apparatus **200** for offering information, that is, the information processing apparatus **200** performs a polling on the server **5** (T**22**). The query is accompanied by information indicating a time of a previous query. When offering information not transmitted to the information processing apparatus **200** is present, that is, new offering information registered after a time indicated by the accompanied information is present, the server **5** that receives the query transmits the new offering information to the information processing apparatus **200** (T**23**). Therefore, the information processing apparatus **200** that receives the offering information stores the received offering information in the data storage unit **35** which is accessible by the device control program **34** (T**24**).
[0079] A polling interval for the server **5** may be the same as the polling interval for each device **1** performed by the device monitoring program **33**, or the polling intervals for the device **1** and the server **5** may be different. For example, the polling interval for the server **5** may be set longer than the polling interval for the device **1**. Further, the polling interval for the server **5** may be changed based on circumstances such as a season or presence or absence of an event. Further, the polling interval may be a polling interval set by the user on the setting screen **80** described later.
[0080] Next, the device monitoring program **33** searches for offering information stored in the data storage unit **35**, and when newly arrived offering information is present, the device monitoring program **33** displays the offering information screen **53** as illustrated in FIG. **3**C as a pop-up on the first home screen **42**A based on the newly arrived offering information (T**25**). That is, similarly to the first status information screen **50** related to the status information, the device monitoring program **33** displays the offering information screen **53** related to the offering information as a pop-up.
[0081] In this processing, when no newly arrived offering information is present, the device monitoring program **33** may display the offering information screen **53** as a pop-up based on latest offering information. Further, when no newly arrived offering information is present and the first status information screen **50** is displayed, the device monitoring program **33** may search for offering information stored in the data storage unit **35** based on the model name displayed in the model name display field **50**C on the first status information screen **50**, and may display the offering information screen **53** as a pop-up on the first home screen **42**A based on the extracted offering information.
Display of Notification List
[0082] Next, in a state in which the information processing apparatus **200** displays the status information screen **50** or displays the offering information screen **53**, when the user clicks a notification list button portion displayed in the notification list button display field **50**G on the first status information screen **50** or the notification list button display field **53**B on the offering information screen **53** with the mouse **41** (T**31**), the device monitoring program **33** instructs the device control program **34** to activate and display a notification list screen **54** (T**32**).
[0083] Then, as illustrated in FIG. **6**, the activated device control program **34** displays all notifications for the monitored device **1** on the display **42** as the notification list screen **54**, based

on the notification information stored in the data storage unit **35** (T**33**).

[0084] The notification list screen **54** displays notification objects **54**A to **54**D. Further, each of the notification objects **54**A to **54**D displays a notification date, a notification message, a target model name, and the like. Then, when a click operation with the mouse **41** on each of the objects **54**A to **54**D is received, a detailed information screen or a reference information screen is displayed.

[0085] The device control program **34** refers to the offering information stored in the data storage unit **35** by the device monitoring program **33** in T**24**, and displays the notification list screen **54** based on the referred offering information. In FIG. **8** described later, even when the user clicks a “notification list” tab **83**A portion with the mouse **41**, the notification list screen **54** can be displayed on the setting screen **80**.

[0086] Further, in a state in which the device monitoring program **33** displays the first status information screen **50**, when the user clicks a character portion or an illustration portion displayed in the offering information display field **50**F with the mouse **41** (T**34**), the device monitoring program **33** instructs the device control program **34** to activate and display a detailed information screen or a reference information screen (not illustrated) (T**35**). Then, the activated device control program **34** displays the detailed information screen and the reference information screen on the display **42** based on the notification information stored in the data storage unit **35** (T**36**).

[0087] In a state in which the offering information screen **53** is displayed, when the user clicks a character portion or an illustration portion displayed in the offering information display field **53**A with the mouse **41** (T**34**), the device monitoring program **33** instructs the device control program **34** to activate and display a detailed information screen or a reference information screen (not illustrated) (T**35**). The activated device control program **34** displays the detailed information screen and the reference information screen on the display **42** based on the notification information stored in the data storage unit **35** (T**36**).

[0088] As described above, in the present embodiment, since the notification information for all the devices **1** monitored by the device monitoring program **33** is stored in the data storage unit **35**, and the device control program **34** displays the notification list screen **54**, the detailed information screen, and the reference information screen on the display **42** based on the notification information stored in the data storage unit **35**, the user can properly understand the offering information currently provided to the device **1**.

[0089] When a function of displaying a screen corresponding to the notification list screen **54** is implemented in the resident device monitoring program **33**, a large amount of resources is consumed for that screen, regardless of whether the screen is being displayed while the device monitoring program **33** is activated. However, in the present embodiment, since the non-resident device control program **34** allows the device monitoring program **33** to display the notification list screen **54**, and does not allow the resident device monitoring program **33** to implement the function of displaying the screen corresponding to the notification list screen **54**, the amount of resources used by the resident device monitoring program **33** can be reduced.

[0090] In the present embodiment, the device monitoring program **33** is configured, in response to a display instruction, to display the notification list screen **54**, the detailed information screen, the reference information screen, and the like on the display **42** based on the notification information stored in the data storage unit **35**. However, the device monitoring program **33** may also be configured, in response to a display instruction (T**31**, T**34**), to access the server **5** and display the notification list screen **54**, the detailed information screen, the reference information screen, and the like on the display **42** based on latest notification information downloaded therefrom (T**33**, T**36**). With this configuration, the user can view the latest notification list screen **54**, the detailed information screen, the reference information screen, and the like related to the device **1**, and can prevent an erroneous operation based on old information.

Various Setting Processing

[0091] Next, FIG. **7** is a sequence diagram illustrating an example of a procedure for various setting processing implemented by the device monitoring program **33** and the device control program **34** of the information processing apparatus **200**, including operations of the device **1**. Hereinafter, details of the various setting processing will be described with reference to FIG. **7**.

[0092] Each processing step in the present embodiment represents processing of the CPU **21** according to an instruction described in a program such as the device monitoring program **33** of the information processing apparatus **200**. The processing by the CPU **21** also includes a hardware control using an API of the OS. In the present specification, a detail description of the OS is omitted, and an operation of each program is described. In addition, in the present specification, for convenience, processing primarily performed by the CPU **21** may be described as being primarily performed by a program or a device.

[0093] That is, as illustrated in FIG. **8**, when the user clicks a box display button **45** portion displayed in a notification area of the taskbar **44** displayed on the display **42** with the mouse **41** to instruct a display of a box (T**41**), the device monitoring program **33** displays a box **60** on the display **42**, and displays, within the box **60**, a first icon **61** and a first badge **62** of the device monitoring program **33**, which is a resident program, respectively (T**42**). The box **60** is an area in which icons of resident programs can be displayed, and can also be referred to as an icon display area.

[0094] The device monitoring program **33** changes a display method of the first badge **62** based on a type of information acquired from the monitored device, for example, the device **1** or the server **5** via the network interface **27**. However, a detail description of the display method of the first badge **62** will be omitted here.

[0095] Next, when the user clicks a first icon **61** portion with the mouse **41** to instruct a display of an activation box **63** (T**43**), the device monitoring program **33** displays the activation box **63** on the display **42** (T**44**).

[0096] In the activation box **63**, characters **63**A “activate control program”, characters **63**B “notification setting”, characters **63**C “notification list”, characters **63**D “device setting”, and the like are displayed, respectively.

[0097] Here, when the user clicks a portion of the characters **63**A “activate control program” in the activation box **63** with the mouse **41** to instruct activation of the device control program **34** (T**45**), the device monitoring program **33** activates the device control program **34** (T**46**).

[0098] The device control program **34** can also be activated by the user clicking an icon **43** portion displayed on the first home screen **42**A with the mouse **41**. Further, a shortcut icon for activating the device control program **34** may be provided in the taskbar **44**.

[0099] When the user clicks a portion of the characters **63**B “notification setting” with the mouse **41**, the device monitoring program **33** instructs the device control program **34** to display a notification setting screen **80**A described later. When the user clicks a portion of the characters **63**C “notification list” with the mouse **41**, the device monitoring program **33** instructs the device control program **34** to display the notification list screen **54**. When the user clicks a portion of characters **63**D “device setting” with the mouse **41**, the device monitoring program **33** instructs the device control program **34** to display a device list screen **85** described later.

[0100] When the device control program **34** is activated, the device control program **34** displays the second home screen **70** of the device control program **34** on the display **42** (T**47**).

[0101] FIG. **9** is a diagram illustrating an example of the display **42** in a state in which the second home screen **70** displayed by the device control program **34** is displayed.

[0102] As illustrated in FIG. **9**, a print icon **71**, a scan icon **72**, a setting button **78**, a “device addition” button **79**, and the like are displayed on the second home screen **70**. The second home screen **70** includes a model name display field **73**, a status information display field **74**, and an ink remaining amount display field **75**.

[0103] The print icon **71** is used when a device to be controlled is caused to perform printing. The scan icon **72** is used when a device to be controlled is caused to perform image reading. The model name display field **73** displays a name of the device **1** to be controlled by the device control program **34**, specifically, a model name of the device **1**.

[0104] The model name display field **73** is configured with a pull-down menu, and when the user clicks a list button **73**A portion with the mouse **41**, a menu (not illustrated) is displayed below the model name display field **73**. The menu displays the model names of all the devices **1** to be monitored by the device monitoring program **33**, that is, the model names and the IP addresses stored in the device table **35**A.

[0105] The print icon **71** and the scan icon **72** are displayed based on a capability of a device selected to be controlled. That is, when the device displayed in the model name display field **73** has a printing function, the print icon **71** is displayed as operable, whereas when the device does not have a printing function, the print icon **71** is not displayed as operable. When the device displayed in the model name display field **73** has an image reading function, the scan icon **72** is displayed as operable, whereas when the device does not have an image reading function, the scan icon **72** is not displayed as operable.

[0106] When the user clicks any one of the model name portions displayed in the list menu with the mouse **41**, the clicked new model name is displayed in the model name display field **73**, and the device **1** to be controlled by the device control program **34** is changed.

[0107] At this time, even if the model name displayed in the model name display field **73** is changed by the user and the device **1** to be controlled by the device control program **34** is changed, the model name displayed in the model name display field **50**C is not changed in a state in which the first status information screen **50** is displayed as a pop-up.

[0108] The model name display field **73** on the second home screen **70** for setting a device **1** to be controlled and the model name display field **50**C for setting a device **1** whose status is checked are often used for different devices **1**, respectively. Therefore, it is more convenient for the user when the device **1** set in the model name display field **73** of the second home screen **70** and the device **1** set in the model name display field **50**C are not linked with each other, as in the present embodiment.

[0109] The status information display field **74** displays the status information on the device **1** displayed in the model name display field **73**. When the device **1** displayed in the model name display field **73** is the printer **1**A, the ink remaining amount display field **75** displays an ink remaining amount of the printer **1**A. When the device control program **34** is activated, as illustrated in FIG. **9**, a second icon **76** and a second badge **77** of the device control program **34** are displayed in the taskbar **44** on the first home screen **42**A (T**48**). The device control program **34** displays, in the second badge **77**, the number of unread offering information among the offering information for all the devices **1** to be monitored by the device monitoring program **33**, which is stored in the data storage unit **35**. That is, the second badge **77** illustrated in FIG. **9** indicates a state in which **20** pieces of unread offering information related to all the devices **1** to be monitored by the device monitoring program **33** are present. The icon **76** displayed in the taskbar **44** is larger in size than the icon **61** displayed in the box **60**. Therefore, the icon **76** can display a badge indicating how many pieces of unread offering information are present.

[0110] Returning to the sequence diagram illustrating an example of the procedure for display processing of the status information and offering information illustrated in FIG. **7**, when the user clicks a setting button **78** portion illustrated in FIG. **9** with the mouse **41** to instruct a display of a setting screen (T**49**), the device control program **34** displays the setting screen **80**, on which the notification setting screen **80**A is displayed, on the display **42** as illustrated in FIG. **10** (T**50**). This processing is an example of the display processing. The setting button **78** is an example of an operation object.

[0111] On the setting screen **80**, an activation timing setting field **81** for setting an activation timing of the device monitoring program **33**, an offering information setting field **82** for setting a type of offering information provided to the user, and the like are displayed. The activation timing setting field **81** displays a first radio button **81**A for always activating the device monitoring program **33** when the information processing apparatus **200** is activated, that is, for setting the device monitoring program **33** to a resident program, and a second radio button **81**B for activating the device monitoring program **33** when the device control program **34** is activated, that is, for setting the device monitoring program **33** to a non-resident program.

[0112] The offering information setting field **82** on the setting screen **80** displays a third radio button **82**A for setting to agree to receive offering information and a fourth radio button **82**B for setting not to receive offering information. Below the third radio button **82**A, a first check box **82**C and a second check box **82**C for selecting a type of offering information to be received are displayed.

[0113] When the user clicks a first radio button **81**A portion with the mouse **41** (T**51**), the device control program **34** executes reception processing for a setting change (T**52**). Thereafter, the device control program **34** passes a setting change notification to the device monitoring program **33** (T**53**). Then, the device monitoring program **33** that receives the setting change notification sets itself to a resident program (T**54**). “Passing a setting change notification to the device monitoring program **33**” may be referred to as “transmitting a setting change notification to the device monitoring program **33**” or “storing a setting change notification in a storage area accessed by the device monitoring program **33**”. The same applies to the description related to T**53** or the like described later.

[0114] When the user clicks a second radio button **81**B portion with the mouse **41** (T**51**), the device control program **34** executes reception processing for the setting change (T**52**). Thereafter, the device control program **34** transmits a setting change notification to the device monitoring program **33** (T**53**). Then, the device monitoring program **33** that receives the setting change notification performs a setting change of the setting information stored in the data storage unit **35** and sets itself to a non-resident program (T**54**).

[0115] When the user clicks a third radio button **82**A portion with the mouse **41** and then clicks the first check box **82**C and a first check box **82**C portion with the mouse **41** to set a type of offering information to be received (T**51**), the device control program **34** executes reception processing for the setting change (T**52**).

[0116] Thereafter, the device control program **34** passes a setting change notification to the device monitoring program **33** (T**53**). Then, the device monitoring program **33** that receives the setting change notification displays offering information in offering information display field **50**F on the first status information screen **50** and the offering information display field **53**A on the offering information screen **53** according to the type of offering information to be received, which is set by the user.

[0117] When the user clicks a fourth radio button **82**B portion with the mouse **41** to set not to receive the offering information (T**51**), the device control program **34** executes reception processing for the setting change (T**52**).

[0118] Thereafter, the device control program **34** passes a setting change notification to the device monitoring program **33** (T**53**). Then, the device monitoring program **33** that receives the setting change notification does not display the offering information in the offering information display field **50**F on the first status information screen **50** and the offering information display field **53**A on the offering information screen **53** according to the setting by the user.

[0119] As described above, in the present embodiment, by setting the type of offering information or the like desired by the user to be downloaded in the offering information setting field **82**, the offering information or the like desired by the user can be provided, and only the offering information desired by the user can be efficiently provided to the user.

[0120] In the present embodiment, the setting screen **80** is configured such that the activation timing of the device monitoring program **33** can also be set, and a monitoring interval time of the device **1** by the device monitoring program **33**, a selection of the device **1** to be monitored, or the like may be set. Further, when there are a plurality of types of notification modes of status information by the device monitoring program **33**, the notification mode may also be set.

[0121] In general, when the resident device monitoring program **33** provides the setting screen **80**, a large amount of resources is consumed for that screen at all times, regardless of whether the screen is being displayed. However, in the present embodiment, since the non-resident device control program **34** displays the setting screen **80** for the resident device monitoring program **33**, and the resident device monitoring program **33** does not provide the setting screen **80**, the amount of resources used by the resident device monitoring program **33** reduced.

[0122] Next, when the user clicks a “managed device list” tab **83**C on the setting screen **80** with the mouse **41** to instruct a display of a device list screen (T**55**), the information processing apparatus **200** (device control program **34**) displays the device list screen **85** on the setting screen **80** based on the model name stored in the device table **35**A stored in the data storage unit **35** (T**56**). A “tab” is a type of object that receives an operation. When a “notification setting” tab **83**B is clicked in a state in which the device list screen **85** is displayed on the setting screen **80**, the device control program **34** displays the notification setting screen **80**A on the setting screen **80**.

[0123] As illustrated in FIG. **11**, the device list screen **85** includes a currently selected model name display field **86** and an other model name display field **87**. The currently selected model name display field **86** displays a model name displayed in the model name display field **73** illustrated in FIG. **9**, that is, a model name and an IP address of the device **1** to be controlled by the device control program **34**. The other model name display field **87** displays model names of the devices **1** to be monitored by the device monitoring program **33**, that is, other model names and IP addresses, excluding the model name displayed in the currently selected model name display field **86**, among the model names stored in the device table **35**A. Further, the other model name display field **87** includes a “delete” button **88** corresponding to each displayed model name.

[0124] Here, when the user clicks any one of “delete” button **88** portions with the mouse **41** to instruct device deletion (T**57**), the device control program **34** executes reception processing for the deletion of the managed device (T**58**). That is, the device control program **34** deletes the model name and the IP address for which the deletion instruction was issued from the list menu displayed below the model name display field **73**.

[0125] At this time, in the present embodiment, since the currently selected model name display field **86** does not include a “delete” button, the user can be prevented from erroneously deleting the model name of the device **1** to be controlled by the device control program **34**.

[0126] Thereafter, the device control program **34** instructs the device monitoring program **33** to delete the managed device (T**59**). Then, the device monitoring program **33** that receives the setting change notification deletes a specific model name and an IP address specified by the model name and the IP address for which the deletion instruction was issued, among the model names stored in the device table **35**A stored in the data storage unit **35** (T**60**). Thereafter, the device monitoring program **33** executes a monitoring operation on the device **1** based on the new device table **35**A excluding the model name for which the deletion instruction was issued (T**61**). That is, this is reflected in the processing in T**13** to T**15** illustrated in FIG. **4**.

[0127] In general, when the resident device monitoring program **33** provides the device list screen **85**, a large amount of resources is consumed for that screen at all times, regardless of whether the screen is being displayed. However, in the present embodiment, since the non-resident device control program **34** displays the device list screen **85**, and the resident device monitoring program **33** does not provide the device list screen **85**, the amount of resources used by the resident device monitoring program **33** can be reduced.

[0128] Next, when the user clicks an “add device” button **79** portion displayed on the second home screen **70** illustrated in FIG. **9** with the mouse **41** to instruct a display of a device addition screen (T**62**), the device control program **34** displays a device addition screen **90** on the display **42** (T**63**).

[0129] As illustrated in FIG. **12**, the device addition screen **90** includes an additional device display field **91**, and displays a “device search” button **92**, an “OK” button **93**, and the like. The additional device display field **91** displays a list of model names and IP addresses of all the devices **1** connected to the information processing apparatus **200**.

[0130] Here, when the user clicks a “device search” button **92** portion with the mouse **41**, the device control program **34** searches for the device **1** connected to the information processing apparatus **200** and then displays the model names of all the searched devices in the additional device display field **91**.

[0131] When the user clicks any one of the model names displayed in the additional device display field **91** with the mouse **41** and then clicks an “OK” button **93** with the mouse **41** to instruct device addition (T**64**), the device control program **34** executes reception processing for the device addition (T**65**). That is, the device control program **34** adds the added model name and IP address to the list menu displayed below the model name display field **73**.

[0132] Thereafter, the device control program **34** instructs the device monitoring program **33** to add a device (T**66**). Then, the device monitoring program **33** that receives a device addition notification adds a model name and an IP address of the device for which the addition instruction was issued, to the device table **35**A stored in the data storage unit **35** (T**67**). Thereafter, the device monitoring program **33** executes a monitoring operation on the device **1** based on the new device table **35**A to which the new model name and IP address are added (T**68**). That is, this is reflected in the processing in T**13** to T**15** illustrated in FIG. **4**.

[0133] In general, when the resident device monitoring program **33** provides the device addition screen **90**, a large amount of resources is consumed for that screen at all times, regardless of whether the screen is being displayed. However, in the present embodiment, since the non-resident device control program **34** displays the device addition screen **90**, and the resident device monitoring program **33** does not provide the device addition screen **90**, the amount of resources used by the resident device monitoring program **33** can be reduced.

[0134] As described above in detail, according to the technique disclosed in the present specification, separate from the resident device monitoring program **33** that acquires status information related to a device, by causing the non-resident device control program **34** capable of transmitting an execution instruction to a device to include a setting screen for the device monitoring program **33**, the amount of resources consumed by the device monitoring program **33** can be reduced and resources for displaying a setting screen for making a setting related to the device monitoring program **33** can be efficiently used, compared with a case of causing the device monitoring program **33** to include the setting screen.

[0135] Further, since the setting screen for the resident device monitoring program **33** is separated from the device monitoring program **33**, the amount of resources consumed by the device monitoring program **33** is less likely affected even if the setting screen is complicated. Therefore, a rich setting screen for the device monitoring program **33** can be provided.

[0136] In addition, when the setting screen for the resident device monitoring program **33** supports multiple languages, it is necessary to provide a setting screen for each language. Therefore, if the device monitoring program **33** includes a setting screen, a size of the device monitoring program **33** is likely to increase, and the amount of resources consumed is likely to increase. By separating the setting screen for the device monitoring program **33** from the device monitoring program **33**, the effect of reducing the amount of resources consumed is remarkable.

[0137] By providing a program set including the resident device monitoring program **33** and the non-resident device control program **34** as one package, the non-resident device control program **34** can be instructed to execute image processing, and the amount of resources consumed by the resident device monitoring program **33** can be reduced.

[0138] The program set including the device management program **33** and the device control program **34** may not be provided as one package. For example, the device monitoring program **33** and the device control program **34** may be provided as separate programs, and after both programs are installed in the OS **31**, both programs may be used as a program set to perform the above cooperative processing.

[0139] The present embodiment is merely an example, and does not limit the present disclosure. Therefore, various improvements and modifications can be naturally made to the present disclosure without departing from the scope thereof. For example, the printer **1**A may not include the operation panel **13**. The information processing apparatus **200** is not limited to the non-volatile memory **24**, and may include any type of large-capacity storage device.

[0140] Further, in the processing in T**12**, when the information processing apparatus **200** receives highly urgent status information such as a sheet jam, the information processing apparatus **200** may be configured to immediately display any one of the first to third status information screens **50** to **52** on the display **42** based on the received status information to notify the user.

[0141] The communication method between the printer **1**A and the information processing apparatus **200** is not limited to wireless communication conforming to the Wi-Fi (registered trademark) standard. For example, the communication method may be wired communication using a USB cable or wireless communication based on other standards such as Bluetooth (registered trademark). Further, a plurality of communication functions may be provided.

[0142] The processing disclosed in the embodiment may be executed by hardware such as a single CPU, a plurality of CPUs, an ASIC, or a combination thereof. In addition, the processing disclosed in the embodiment can be implemented in various modes such as a recording medium in which a program for executing the processing is recorded, or a method.

Claims

1. A non-transitory computer readable storage medium storing a program set, the program set comprising a first program and a second program, the first program being a resident program, the second program being a non-resident program, wherein the second program, when executed by a processor of an information processing apparatus, causes the information processing apparatus to perform second operations, the second operations comprising: transmitting, to an image processing device via a communication device of the information processing apparatus, an execution instruction for causing a job related to image processing to be executed; and displaying a setting screen configured to receive a setting related to the first program, and wherein the first program, when executed by the processor, causes the information processing apparatus to perform first operations, the first operations comprising: communicating, in accordance with the setting received on the setting screen displayed in the second operations, with the image processing device via the communication device; acquiring information related to the image processing device through a communication with the image processing device; and displaying notification information based on the information related to the image processing device acquired in the acquiring the information related to the image processing device.

2. The non-transitory computer readable storage medium according to claim 1, wherein the first operations comprise: displaying an icon corresponding to the first program in a predetermined area in which one or more icons of respective one or more resident programs are allowed to be displayed; and activating the second program to execute the displaying the setting screen in response to a predetermined operation received via the icon corresponding to the first program.

3. The non-transitory computer readable storage medium according to claim 1, wherein the second operations comprise: displaying a screen in which an operation object is displayed, the operation object being configured to receive a display instruction of the setting screen; and executing the displaying the setting screen in response to receipt of an operation on the operation object.

4. The non-transitory computer readable storage medium according to claim 1, wherein the setting related to the first program, received on the setting screen displayed in the second operations, includes a setting as to whether to make the first program resident, wherein in a case in which a setting to make the first program resident is made, the first program is activated upon activation of the information processing apparatus and is resident, and wherein in a case in which a setting not to make the first program resident is made, the first program is not activated upon activation of the information processing apparatus.

5. The non-transitory computer readable storage medium according to claim 1, wherein in the acquiring the information related to the image processing device in the first operations, a status of the image processing device is acquired as the information related to the image processing device, and wherein in the displaying the notification information in the first operations, status information based on the status of the image processing device is displayed as the notification information.

6. The non-transitory computer readable storage medium according to claim 5, wherein the first program, when executed by the processor, allows a plurality of image processing devices to be set as a monitoring target, wherein the setting received on the setting screen displayed in the second operations includes a setting of an image processing device to be set as the monitoring target, wherein in the displaying the setting screen in the second operations, an adding operation for adding an image processing device to the monitoring target is allowed to be received via the setting screen, and wherein the acquiring the information related to the image processing device in the first operations comprise, in a case in which a first image processing device is added to the monitoring target by the adding operation, starting communication, via the communication device, with the first image processing device added to the monitoring target, and acquiring the status from the first image processing device.

7. The non-transitory computer readable storage medium according to claim 6, wherein in the displaying the setting screen in the second operations, an excluding operation for excluding an image processing device from the monitoring target is allowed to be received via the setting screen, and wherein the acquiring the information related to the image processing device in the first operations comprise, in a case in which a second image processing device is excluded from the monitoring target by the excluding operation, stopping communication, via the communication device, with the second image processing device excluded from the monitoring target.

8. The non-transitory computer readable storage medium according to claim 6, wherein the first operations further comprise displaying a first selection screen configured to receive a selection of one of image processing devices set as the monitoring target, and the displaying the notification information in the first operations comprises displaying the status information on the one of image processing devices selected on the first selection screen, wherein the second operations further comprise displaying a second selection screen configured to receive a selection of one of image processing devices set as the monitoring target, and the transmitting the execution instruction comprises transmitting, to the one of image processing devices selected on the second selection screen via the communication device, the execution instruction for causing the job related to image processing to be executed, wherein in the first operations, the one of image processing devices selected on the first selection screen is not changed even in a case in which the one of image processing devices selected on the second selection screen in the second operations is changed, and wherein in the second operations, the one of image processing devices selected on the second selection screen is not changed even in a case in which the one of image processing devices selected on the first selection screen in the first operations is changed.

9. The non-transitory computer readable storage medium according to claim 1, wherein the acquiring the information related to the image processing device in the first operations comprises: communicating, in accordance with the setting received on the setting screen displayed in the second operations, also with a server via the communication device, the server being configured to provide service-related information related to a service available to the image processing device; and acquiring the service-related information as the information related to the image processing device through a communication with the server, and wherein the displaying the notification information in the first operations comprises displaying the service-related information as the notification information.

10. The non-transitory computer readable storage medium according to claim 9, wherein the first operations further comprise storing, in a memory of the information processing apparatus, the service-related information acquired in the acquiring the service-related information, the memory being configured to store a plurality of pieces of the service-related information, and wherein the second operations further comprise: reading the service-related information stored in the memory; and displaying a list of the service-related information read from the memory.

11. A non-transitory computer readable storage medium storing a program set, the program set comprising an acquisition program and a display program, the acquisition program being a resident program, the display program being a non-resident program, wherein the acquisition program, when executed by a processor of an information processing apparatus, causes the information processing apparatus to perform first operations, the first operations comprising: communicating with a server via a communication device of the information processing apparatus, the service-related information being related to a service available to an image processing device, the server being configured to provide the service-related information; acquiring the service-related information through a communication with the server; and storing, in a memory of the information processing apparatus, the service-related information acquired in the acquiring the service-related information, the memory being configured to store a plurality of pieces of the service-related information, and wherein the display program, when executed by the processor, causes the information processing apparatus to perform second operations, the second operations comprising: reading the service-related information stored in the memory; and displaying a list of the service-related information read from the memory.

12. An information processing apparatus comprising: a communication device; a processor; and a memory storing a program set, the program set comprising a first program and a second program, the first program being a resident program, the second program being a non-resident program, wherein the second program, when executed by the processor, causes the information processing apparatus to perform second operations, the second operations comprising: transmitting, to an image processing device via the communication device, an execution instruction for causing a job related to image processing to be executed; and displaying a setting screen configured to receive a setting related to the first program, and wherein the first program, when executed by the processor, causes the information processing apparatus to perform first operations, the first operations comprising: communicating, in accordance with the setting received on the setting screen displayed in the second operations, with the image processing device via the communication device; acquiring information related to the image processing device through a communication with the image processing device; and displaying notification information based on the information related to the image processing device acquired in the acquiring the information related to the image processing device.